

Morgan Huberty

[Email](#) | [LinkedIn](#) | [Website](#) - <https://mhubert3.github.io/>

Education

Johns Hopkins University

Expected May 2027

Bachelor of Science in Computer Engineering

Relevant Coursework: Data Structures, Computer Fundamentals, Signals & Systems, FPGA Lab, Mastering Electronics

Software Projects

Big Integers

C++

- Implemented a BigInt class (vector<uint64_t> magnitude, bool sign) supporting arbitrary-precision add, sub, mul, div, bit ops, and hex/dec conversion, with comprehensive unit tests

Key/Value Store

C++

- Built a Redis-like in-memory K/V store with custom message serialization, multithreaded server (auto-commit & transactions with table-level locking), client utilities, and unit tests

Cache Simulator

C, C++

- Simulated a configurable set-associative cache (size, block, LRU/FIFO, write policies) that outputs hit/miss, cycle counts from benchmark traces

DTMF Singla Decoder

MATLAB

- Decoded DTMF digit sequences using FFT-based tone detection and envelope segmentation

Hardware Projects

Stopwatch Controller

VHDL

- Built a multi-bit synchronous counter with start/stop toggle and reset logic driven by debounced button inputs
- Verified functionality on hardware; adjusting timing, control behavior to resolve metastability, synchronization issues

I²C Serial Interface

VHDL

- Implemented an I²C master with start/stop conditions, ACK/NACK handling, and open-drain SDA/SCL signaling
- Tested data transfer with a PIC microcontroller and debugged timing behavior using incremental hardware validation

Experience

#iVoted Impact Concerts at SNF Agora Institute

May 2025 - Present

Data Science Researcher

- Analyzed music-listener and voter-demographic datasets (RedistrictingDataHub, Geocorr APIs) to inform #iVoted interventions
- Built projection dashboard for voter registrations using Python, Excel, and tracked data from #iVoted interventions

National Society of Black Engineers

September 2023 - Present

Vice President; previous Finance Chair

- Managed financial planning for chapter activities, ensuring allocating and stewarding a \$9K annual budget
- Developed marketing materials to promote member engagement and outreach by leveraging digital platforms
- Coordinated fundraising and sponsorship efforts for \$3500 worth of income

Skills

Programming Languages: C, C++, Python (NumPy, Matplotlib, Pandas), MATLAB, VHDL, Bash

Tools: LTSpice, Xilinx Vivado, Git, Linux (Ubuntu), Docker, Vercel, Tinkercad, Fusion360, Excel, Slack, Canva

Familiarity with KiCad, AWS, Google Cloud, PyTorch, TensorFlow, x86-64 Assembly, Digital Logic & Hardware